

When Online Bayes Meets Expert Advice

Following the Best Bayesian Learner

Jungbin Jun

Department of Statistics, Inha University

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Motivation

- Online learning studies how to make predictions while updating a model from sequentially arriving data.
- At each round t , the learner chooses $\hat{\theta}_t \in \Theta$.
- After committing to this choice, the learner incurs a loss $\ell_t(\hat{\theta}_t)$ and updates for the next round.

Observe $x_t \rightarrow$ Choose $\hat{\theta}_t \rightarrow$ Suffer $\ell_t(\hat{\theta}_t) \rightarrow$ Observe $x_{t+1} \rightarrow \dots$

- Bayesian learning can also be viewed as an online learning procedure.
- Starting from a prior $\pi(\theta)$, we sequentially update the distribution using the current posterior as the next prior.
- Under generalized Bayes,

$$p_{t+1}(\theta) \propto \exp(-\eta \ell_t(\theta)) p_t(\theta).$$

- A natural decision is the posterior mean

$$\hat{\theta}_{t+1} = \mathbb{E}_{\theta \sim p_{t+1}}[\theta].$$

Why expert advice?

- Online learning requires tuning without a validation loop.
- In online Bayes, the prior, temperature, and update rule can strongly affect performance.
- Under distribution shift, the data distribution may evolve over time, so a single Bayesian update can be unreliable.
- Expert-advice frameworks are widely used in online learning to hedge against this kind of sensitivity [Gradu et al., 2023; Gibbs and Candès, 2024; Angelopoulos et al., 2025].
- We bring this framework into online Bayesian updating.

Proposed Method

- We maintain multiple K Bayesian experts in parallel.
- Expert $k \in [K]$ is specified by the prior π and temperature η .
- Expert k updates its own posterior

$$p_{t+1}^{(k)}(\theta) \propto \exp\left(-\eta^{(k)} \sum_{i=1}^t \ell_i(\theta)\right) \pi^{(k)}(\theta),$$

and predicts by its posterior mean

$$\hat{\theta}_t^{(k)} = \mathbb{E}_{\theta \sim p_t^{(k)}}[\theta].$$

Online Bayes with Expert Advice

Setup. For each $k \in [K]$, we assign an aggregation weight $w_{t,k}$ to expert k . At $t = 1$, initialize $w_1 \in \Delta_K$ (e.g., discrete uniform).

Weight update.

$$\tilde{w}_{t+1,k} \propto w_{t,k} \exp(-\gamma \ell_t(\hat{\theta}_t^{(k)})) \quad \text{Loss-based reweighting}$$

$$w_{t+1,k} \propto (1 - \sigma) \tilde{w}_{t+1,k} + \sigma \frac{1}{K} \quad \text{Prevent collapse via sharing}$$

Aggregation.

$$\hat{\theta}_{t+1} = \sum_{k \in [K]} w_{t+1,k} \hat{\theta}_{t+1}^{(k)}$$

Summary of proposed algorithm

Algorithm 1 Online Bayes with Expert Aggregation

- 1: **Experts:** $\mathcal{E}_k = (\pi^{(k)}, \eta^{(k)}), k \in [K]$
 - 2: **Initialize:** $p_1^{(k)} \leftarrow \pi^{(k)}$ for all k , and $w_1 \in \Delta_K$
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Compute $\hat{\theta}_t^{(k)} \leftarrow \mathbb{E}_{\theta \sim p_t^{(k)}}[\theta]$
 - 5: Predict $\hat{\theta}_t \leftarrow \sum_{k=1}^K w_{t,k} \hat{\theta}_t^{(k)}$
 - 6: Suffer $\ell_t(\hat{\theta}_t^{(k)})$
 - 7: Reweight $\tilde{w}_{t+1,k} \propto w_{t,k} \exp(-\gamma \ell_t(\hat{\theta}_t^{(k)}))$
 - 8: Share $w_{t+1,k} \leftarrow (1 - \sigma) \tilde{w}_{t+1,k} + \sigma / K$
 - 9: Update $p_{t+1}^{(k)}(\theta) \propto \exp(-\eta^{(k)} \ell_t(\theta)) p_t^{(k)}(\theta)$
 - 10: **end for**
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Approximating Online Bayes

Optimization View of Online Bayes

- By the variational formula [[Donsker and Varadhan, 1976](#)],

$$p_{t+1}(\theta) = \arg \min_{p \in \mathcal{P}(\Theta)} \left\{ \mathbb{E}_{\theta \sim p} \left[\sum_{i=1}^t \ell_i(\theta) \right] + \frac{1}{\eta} \text{KL}(p \parallel \pi) \right\}.$$

- Hence, online Bayes can be viewed as a KL-regularized online optimization problem.
- This perspective naturally motivates variational approximation.

Approximating Online Bayes : Online Variational Inference

- In practice, optimizing over the full distribution class $\mathcal{P}(\Theta)$ may be intractable.
- Instead, we restrict the posterior to a tractable variational family $\{q_\phi : \phi \in \Phi\}$:

$$\phi_{t+1} = \arg \min_{\phi \in \Phi} \left\{ \mathbb{E}_{\theta \sim q_\phi} \left[\sum_{i=1}^t \ell_i(\theta) \right] + \frac{1}{\eta} KL(q_\phi \parallel \pi) \right\}$$

Approximating Online Bayes : Sequential Variational Approximation

- We further linearize the instantaneous variational loss around the current iterate:

$$\mathbb{E}_{\theta \sim q_\phi}[\ell_t(\theta)] =: \bar{L}_t(\phi) \approx \bar{L}_t(\phi_t) + (\phi - \phi_t)^\top \nabla \bar{L}_t(\phi_t).$$

- The resulting update is **Sequential Variational Approximation** [Chérif-Abdellatif et al., 2019]:

$$\phi_{t+1} = \arg \min_{\phi \in \Phi} \left\{ \sum_{i=1}^t \phi^\top \nabla \bar{L}_i(\phi_i) + \frac{1}{\eta} KL(q_\phi \parallel \pi) \right\}.$$

Theoretical Analysis

A Goodness metric : Regret

Definition

$$\text{Regret} = \underbrace{\sum_{t=1}^T \ell_t(\hat{\theta}_t)}_{\text{Our Choice}} - \underbrace{\inf_{\theta \in \Theta} \sum_{t=1}^T \ell_t(\theta)}_{\text{Oracle Performance}}$$

- Goal: achieve sublinear regret bounds (e.g., $O(\log T)$, $O(\sqrt{T})$, ...)
- Why is sublinear regret desirable?

$$\frac{\text{Regret}}{T} \rightarrow 0$$

- In other words, under no regret, we asymptotically perform as well as the oracle.

Regret Bound for Online Bayes

Theorem

Assume $0 \leq \ell_t(\theta) \leq B$ and ℓ_t is convex. Then

$$\begin{aligned} \sum_{t=1}^T \ell_t(\hat{\theta}_t) &\leq \inf_{p \in \mathcal{P}(\Theta)} \left\{ \mathbb{E}_{\theta \sim p} \left[\sum_{t=1}^T \ell_t(\theta) \right] + \underbrace{\min_{k \in [K]} \left(\frac{\eta^{(k)} B^2 T}{8} + \frac{KL(p \parallel \pi^{(k)})}{\eta^{(k)}} \right)}_{\text{best expert term}} \right\} \\ &\quad + \underbrace{\gamma KB^2 + \frac{1}{\gamma} (\log(K/\sigma) + 2T\sigma)}_{\text{tracking overhead}}. \end{aligned}$$

- With suitably chosen time-dependent $\eta^{(k)}$, γ , and σ and together with additional assumption on KL divergence, the regret becomes sublinear.

$$\text{Regret} := \sum_{t=1}^T \ell_t(\hat{\theta}_t) - \inf_{p \in \mathcal{P}(\Theta)} \left\{ \mathbb{E}_{\theta \sim p} \left[\sum_{t=1}^T \ell_t(\theta) \right] \right\} \leq \tilde{O}(\sqrt{T}).$$

Regret Bounds for Sequential Variational Approximation

Theorem

Assume $\mathbb{E}_{\theta \sim q_\phi}(\theta)$ is L -Lipschitz with respect to ϕ and convex and the KL divergence $\phi \mapsto KL(q_\phi, q_{\phi_1})$ is α -strongly convex. Then

$$\begin{aligned} \sum_{t=1}^T \ell_t(\hat{\theta}_t) \leq & \inf_{\phi \in \Phi} \left\{ \mathbb{E}_{\theta \sim q_\phi} \left[\sum_{t=1}^T \ell_t(\theta) \right] + \underbrace{\min_{k \in [K]} \left(\frac{\eta^{(k)} L^2 T}{\alpha} + \frac{KL(q_\phi \| \pi^{(k)})}{\eta^{(k)}} \right)}_{\text{best expert term}} \right\} \\ & + \underbrace{\gamma KB^2 + \frac{1}{\gamma} (\log(K/\sigma) + 2T\sigma)}_{\text{tracking overhead}}. \end{aligned}$$

Numerical Experiments

Online Regression in Non-Stationary Settings

- Datasets:
 - **Elevators** [[Torgo, 2024](#)]: covariates are aircraft control-related features, and the target is an elevator-control value.
 - **SARCOS** [[Rasmussen, 2003](#)]: covariates are 21 robot-arm kinematic variables (joint positions, velocities, accelerations), and the target is the corresponding joint torque.

Online Regression in Non-Stationary Settings

- Baseline expert: Gaussian process (GP)
- For efficient online computation, we use Dynamic Online Ensembles of Basis Expansions (DOEBE) [[Waxman and Djuric, 2024](#)] as a tractable online surrogate for GP regression.
- For each expert, we use the negative log-likelihood loss:

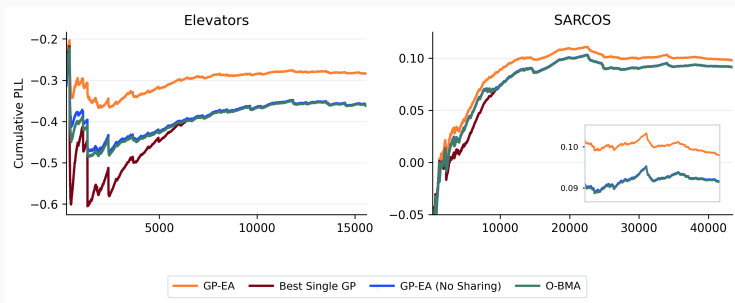
$$\ell_t^{(k)}(\theta) = -\log p(y_t \mid x_t, \theta).$$

- The expert grid is constructed over prior variances, with a fixed learning rate $\eta = 1$:

$$\sigma_k^2 \in \{10^{-3}, 10^{-2}, 10^{-1}, 10^0, 10^1\}.$$

Results

- Compared with other ensemble methods, such as GP-EA (No Sharing) for ablation and Online Bayesian Model Averaging (O-BMA) [Waxman, Llorente, et al., 2026].



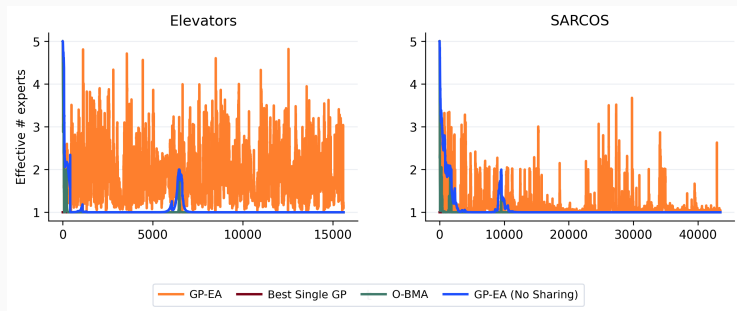
- GP-EA achieves stronger cumulative predictive performance than non-sharing and O-BMA baselines.

Results

- Consider the effective number of experts:

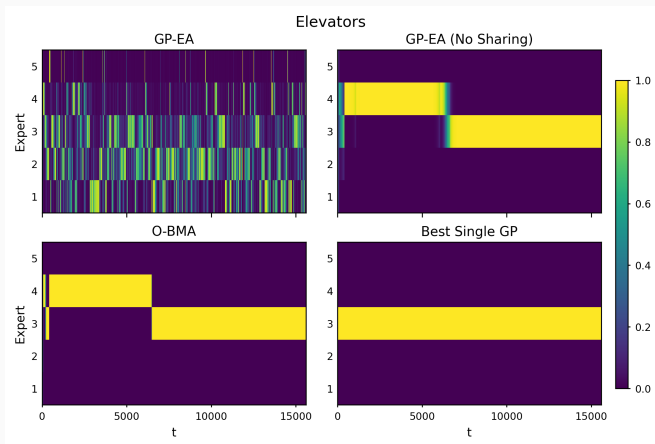
$$N_{\text{eff}} = \frac{1}{\sum_{k \in [K]} w_k^2}.$$

- Uniform $\Rightarrow N_{\text{eff}} = K$; concentrated $\Rightarrow N_{\text{eff}} = 1$.



- GP-EA keeps multiple experts active; no-sharing baselines collapse quickly.

Results



GP-EA adapts without collapsing to a single expert.

Results



GP-EA adapts without collapsing to a single expert.

Takeaways

- We propose online Bayes with expert advice.
- The method tracks the best Bayesian learner with regret guarantees.
- In experiments, weight sharing improves adaptivity over non-sharing baselines.

Thank you!

Questions?

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